

4 Dynamics of a particle moving in a straight line or plane

What students need to learn:

The concept of a force. Newton's laws of motion.

Simple problems involving constant acceleration in scalar form or as a vector of the form $a\mathbf{i} + b\mathbf{j}$.

Simple applications including the motion of two connected particles.

Problems may include

- (i) the motion of two connected particles moving in a straight line or under gravity when the forces on each particle are constant; problems involving smooth fixed pulleys and/or pegs may be set;
- (ii) motion under a force which changes from one fixed value to another, eg a particle hitting the ground;
- (iii) motion directly up or down a smooth or rough inclined plane.

Momentum and impulse. The impulse-momentum principle. The principle of conservation of momentum applied to two particles colliding directly.

Knowledge of Newton's law of restitution is not required. Problems will be confined to those of a one-dimensional nature.

Coefficient of friction.

An understanding of $F = \mu R$ when a particle is moving.

5 Statics of a particle

What students need to learn:

Forces treated as vectors. Resolution of forces.

Equilibrium of a particle under coplanar forces.
Weight, normal reaction, tension and thrust, friction.

Only simple cases of the application of the conditions for equilibrium to uncomplicated systems will be required.

Coefficient of friction.

An understanding of $F \leq \mu R$ in a situation of equilibrium.

6 Moments

What students need to learn:

Moment of a force.

Simple problems involving coplanar parallel forces acting on a body and conditions for equilibrium in such situations.

Unit M2 Mechanics 2

GCE Mathematics, GCE AS and GCE Further Mathematics, GCE AS and GCE Further Mathematics (Additional) A2 optional unit

M2.1 Unit description

Kinematics of a particle moving in a straight line or plane; centres of mass; work and energy; collisions; statics of rigid bodies.

M2.2 Assessment information

Prerequisites

A knowledge of the specification for M1 and its prerequisites and associated formulae, together with a knowledge of algebra, trigonometry, differentiation and integration, as specified in C1, C2 and C3, is assumed and may be tested.

Examination

The examination will consist of one 1½ hour paper. It will contain about seven questions with varying mark allocations per question which will be stated on the paper. All questions may be attempted.

Calculators

Students are expected to have available a calculator with at least the following keys: $+$, $-$, $\times$, $\div$, π , x^2 , $\sqrt{x}$, $\frac{1}{x}$, x^y , $\ln x$, e^x , sine, cosine and tangent and their inverses in degrees and decimals of a degree, and in radians; memory. Calculators with a facility for symbolic algebra, differentiation and/or integration are not permitted.

Formulae

Students are expected to know any other formulae which might be required by the specification and which are not included in the booklet, *Mathematical Formulae including Statistical Formulae and Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage.

Students will be expected to know and be able to recall and use the following formulae:

$$\text{Kinetic energy} = \frac{1}{2}mv^2$$

$$\text{Potential energy} = mgh$$

1 Kinematics of a particle moving in a straight line or plane

What students need to learn:

Motion in a vertical plane with constant acceleration, eg under gravity.

Simple cases of motion of a projectile.

Velocity and acceleration when the displacement is a function of time.

Differentiation and integration of a vector with respect to time.

The setting up and solution of

equations of the form $\frac{dx}{dt} = f(t)$ or $\frac{dv}{dt} = g(t)$ will be consistent with the level of calculus in C2.

For example, given that,

$\mathbf{r} = t^2\mathbf{i} + t^{3/2}\mathbf{j}$, find $\dot{\mathbf{r}}$ and $\ddot{\mathbf{r}}$ at a given time.

2 Centres of mass

What students need to learn:

Centre of mass of a discrete mass distribution in one and two dimensions.

Centre of mass of uniform plane figures, and simple cases of composite plane figures.

The use of an axis of symmetry will be acceptable where appropriate. Use of integration is not required. Figures may include the shapes referred to in the formulae book. Results given in the formulae book may be quoted without proof.

Simple cases of equilibrium of a plane lamina.

The lamina may

- (i) be suspended from a fixed point;
- (ii) free to rotate about a fixed horizontal axis;
- (iii) be put on an inclined plane.

3 Work and energy

What students need to learn:

Kinetic and potential energy, work and power. The work-energy principle. The principle of conservation of mechanical energy.

Problems involving motion under a constant resistance and/or up and down an inclined plane may be set.

4 Collisions

What students need to learn:

Momentum as a vector. The impulse-momentum principle in vector form. Conservation of linear momentum.

Direct impact of elastic particles. Newton's law of restitution. Loss of mechanical energy due to impact.

Students will be expected to know and use the inequalities $0 \leq e \leq 1$ (where e is the coefficient of restitution).

Successive impacts of up to three particles or two particles and a smooth plane surface.

Collision with a plane surface will not involve oblique impact.